

Flask Practical – All Common Commands

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Objective

To install, create, run, test, and manage a Flask application using Python.

1. Check Python and pip

```
python --version
py --version
pip --version
```

2. Create a Virtual Environment

```
python -m venv venv
venv\Scripts\activate
```

3. Upgrade pip

```
python -m pip install --upgrade pip
```

4. Install Flask

```
pip install flask
```

5. Install Optional Useful Packages

```
pip install requests
pip install flask-cors
pip install python-dotenv
```

6. Create the Application File

```
from flask import Flask, jsonify, request

app = Flask(__name__)

@app.route('/')
def home():
    return 'Hello, Usharani! Flask is running.'

@app.route('/items/<int:item_id>')
def get_item(item_id):
    return jsonify({'item_id': item_id})

@app.route('/items', methods=['POST'])
def create_item():
    data = request.get_json()
    return jsonify(data), 201

if __name__ == '__main__':
    app.run(debug=True)
```

7. Run the Flask Application

```
python app.py
flask --app app run
flask --app app run --debug
flask --app app run --host 127.0.0.1 --port 5000
```

8. Stop the Server

Press CTRL+C in the terminal.

9. Test in Browser

```
http://127.0.0.1:5000/
http://127.0.0.1:5000/items/10
```

10. Test with curl

```
curl http://127.0.0.1:5000/
curl http://127.0.0.1:5000/items/10
curl -X POST http://127.0.0.1:5000/items -H "Content-Type: application/json" -d
"{\"name\":\"Room\",\"price\":1500}"
```

11. Flask Environment Commands

```
flask --version
flask --help
flask --app app routes
```

12. Freeze and Install Dependencies

```
pip freeze
pip freeze > requirements.txt
pip install -r requirements.txt
```

13. Check Installed Packages

```
pip show flask
pip list
```

14. Uninstall Flask

```
pip uninstall flask
```

15. Common Project Commands

```
mkdir flask_project
cd flask_project
code .
type nul > app.py
```

Result

The Flask application can be installed, created, executed, tested through browser and curl, and managed using common Flask and Python package commands.